

## LAB 1: WHAT IS THE GENETIC MATERIAL?

You all know that DNA is the genetic material, because that is what you have been taught.

Prove it!

Bacterial strain A carries a gene conferring resistance to ampicillin.

Bacterial strain B is sensitive to ampicillin.

There are two kinds of petri dishes: one contains ampicillin, the other does not.

Here are some recipes and protocols.

#### Protein extraction

1. Add 1.5 ml of stationary phase bacterial culture to a microcentrifuge tube.
2. Centrifuge bacteria for 1 min at 12,000rpm.
3. Remove and discard all supernatant.
4. Resuspend pelleted bacteria in 50 ul of resuspension buffer (RBuff).
5. Add 10 ul of 10 mg/ml lysozyme (Lys).
6. Put on ice for 5 min.
7. Add 300 µl of Protein lysis buffer (PrLys) and mix well.
8. Incubate at 65°C for 15 min. Mix well every 5 min.
9. Add 34 ul of 2M Potassium Phosphate (pH 7.4) (PO4Buff). Mix well.
10. Cool on ice for 30 min.
11. Centrifuge for 5 min at 12,000rpm.
12. Carefully remove most of supernatant to a new tube. Discard pellet.
13. Add 1/10 volume of 10X enzyme digestion buffer supplement (EDSup).
14. Add 1ul of DNase and 1 ul of RNase
15. Incubate 1 hour at 37°C.
16. Store.

#### RNA extraction

1. Add 1.5 ml of stationary phase bacterial culture to a microcentrifuge tube.
2. Centrifuge bacteria for 1 min at 12,000rpm.
3. Remove and discard all supernatant.
4. Resuspend pelleted bacteria in 350 ul of RNA lysis buffer (RNA Lys). Mix well.
5. Load lysate onto the clearing filter (purple ring). Centrifuge for 1 min at 12,000 rpm. Discard filter.
6. Add 350 ul of 70% ethanol (70% EtOH). Mix well.
7. Load diluted lysate onto binding filter (light blue ring). Centrifuge for 30 sec at 12,000 rpm.
8. Place filter into new microcentrifuge tube.
9. Add 350 ul of membrane desalting buffer (DSbuff) to the filter.
10. Centrifuge for 1 min at 12,000 rpm.
11. Add 95 ul of DNase reaction mix (DNAgone) to the center of the filter.
12. Incubate at room temp for 15 min.
13. Inactivate the DNase by adding 200 ul of DNase inactivation buffer (DINbuff).
14. Centrifuge for 30 sec at 12,000 rpm and place the filter in a new microcentrifuge tube.
15. Add 600 ul of RNA wash buffer (Wbuff).
16. Centrifuge for 30 sec at 12,000 rpm and place the filter in a new microcentrifuge tube.

17. Add 250  $\mu$ l of RNA wash buffer (Wbuff).
18. Centrifuge for 2 min at 12,000 rpm and place the filter in a new microcentrifuge tube.
19. Elute the RNA by adding 50  $\mu$ l of ultrapure water ( $\text{ddH}_2\text{O}$ ) to the center of the filter.
20. Centrifuge for 1 min at 12,000 rpm g.
21. Discard filter, store tube with eluent.

#### DNA extraction

1. Add 1.5 ml of stationary phase bacterial culture to a microcentrifuge tube.
2. Centrifuge bacteria for 1 min at 12,000 rpm.
3. Remove and discard all supernatant.
4. Resuspend pelleted bacteria in 50  $\mu$ l of resuspension buffer (RBuff).
5. Add 10  $\mu$ l of 10 mg/ml lysozyme (Lys).
6. Put on ice for 5 min.
7. Add 300  $\mu$ l of DNA Lysis buffer (DNA Lys) and mix well.
8. Incubate at 65°C for 15 min. Gently mix every 5 min.
9. Cool the samples to 37°C. Then add 1  $\mu$ l of 5  $\mu$ g/ $\mu$ l RNase A (RNAgone) to the sample and mix well.
10. Incubate at 37°C for 30 min.
11. Place the samples on ice for 3-5 min.
12. Add 150  $\mu$ l of Protein Precipitation Reagent (Hi Salt) and mix by vortexing for 10 sec.
13. Place the samples on ice for 3 minutes.
14. Centrifuge for 10 min at 4°C at top speed.
15. Transfer the supernatant to a new microcentrifuge tube. Discard the pellet.
16. Add 500  $\mu$ l of isopropanol (IsoP) to the supernatant. Mix by inverting tube many times (>20).
17. Centrifugation for 10 min at 4°C at top speed.
18. Remove the isopropanol. Wait 2 min and carefully remove the remaining isopropanol.
19. Rinse the pellet carefully with 70% ethanol (70% EtOH). Remove all ethanol as in step 18.
20. Leave the open tube on the side on the bench to evaporate remaining ethanol.
21. Resuspend the DNA in 50  $\mu$ l of Tris/EDTA Buffer (TE). Store.

#### Opening bacteria to the environment

The target bacteria in this procedure have been chemically treated (made competent) to increase the porosity of the cell wall. Macromolecules can therefore easily move from the outside media into the bacterial cytoplasm.

1. Thaw the competent bacteria on ice (~10min)
2. Add the target material (5  $\mu$ l)
3. Place on ice for 30 min.
4. Place in 42°C water bath for 30 sec
5. Place on ice for 2 min.
6. Add 1 ml LB media (to a final volume of 1ml, and transfer to 50 ml tubes)
7. Allow recovery for 1 hr at 37°C while shaking.
8. Centrifuge 2-5 min at 5,000 rpm
9. Resuspend in 100  $\mu$ l LB
10. Plate and incubate at 37°C overnight.